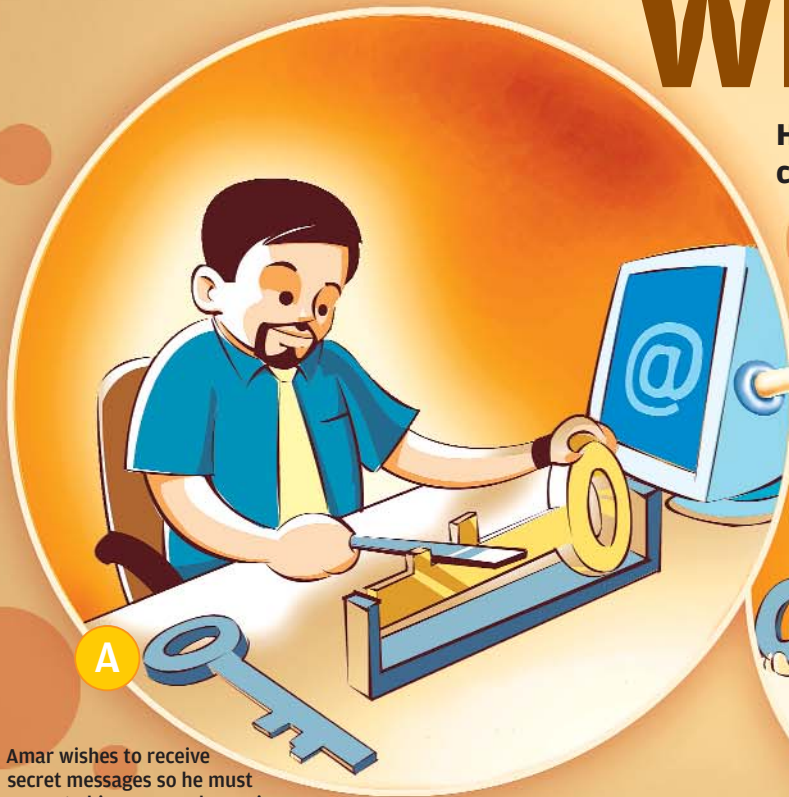


# Write It,

Here's how those super-confidential e-mails are sent



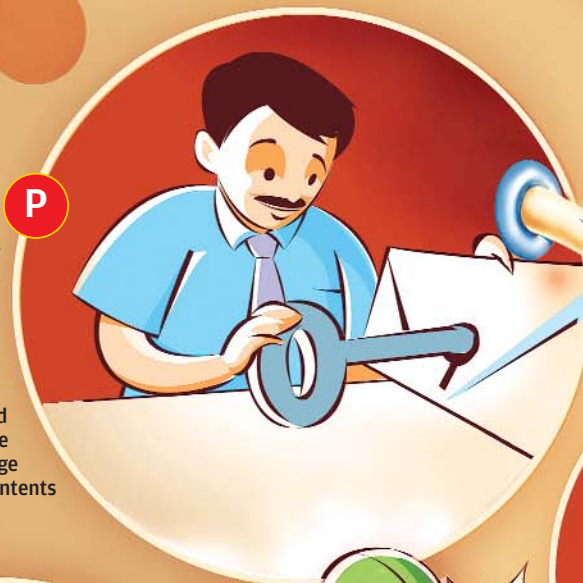
A

**A** Amar wishes to receive secret messages so he must generate his message key pair. This pair has a unique property—a message locked with one key can only be unlocked by the other matched key. The same key cannot be used to lock as well as unlock the secret message. One key (Grey) is his 'Public Key', that he must share with the whole world, while the other (Golden) key is his 'Private Key' that he must keep secret and give to no one.



B

**B** Amar distributes his 'Public Key' by attaching it to email he sends, or on his personal web site, or with a 'Public Key' directory that anyone can look up. They could be Amar's friends or spies who snoop on all the messages he receives.



P

**P** When Amar's colleague Birbal wants to send Amar a secret message, he locks (encrypts) the message using Amar's 'Public Key'. Now the message can be unlocked only using Amar's private key. He sends the message like any other but the contents look like garbage.



Q



X

Y

**X** A Spy is tapping Amar's Internet connection and makes a copy of the message Birbal sends to Amar. The message is encrypted so the Spy tries to use Amar's 'Public Key' to unlock it. But he fails. Once a message is locked with one key, it can only be unlocked by the other key in the pair!